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Yao Fan  ; Yaxuan Duan; Zhengshang Da; Yang Yue 



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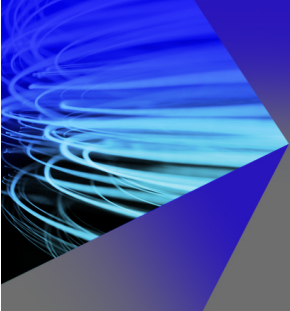
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Yao Fan,^{1,2,3,4,a)} Yaxuan Duan,^{1,3,4} Zhengshang Da,^{1,3,4} and Yang Yue²

AFFILIATIONS

¹Advanced Optical Instrument Laboratory, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China

²School of Information and Communications Engineering, Xi'an Jiaotong University, Xi'an 710049, China

³University of Chinese Academy of Sciences, Xi'an 710119, China

⁴Xi'an Key Laboratory of High Power Laser Measurement Technology and Instrument, Xi'an 710119, China

^{a)}Author to whom correspondence should be addressed: fanyao6227385@stu.xjtu.edu.cn

ABSTRACT

This paper introduces an optimized method for zonal wavefront reconstruction utilizing Fan’s model, specifically tailored to enhance the frequency response. Analysis of the system frequency response demonstrates a 27% increase in bandwidth compared to the Southwell model. Examination of reconstruction errors at various frequency points reveals consistently smaller values when compared to the Southwell model. Validation through numerical simulations and real experiments underscores the superior performance of the proposed reconstructor, particularly noticeable at higher response levels within the mid- and high-frequency domains.

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I. INTRODUCTION

The Shack–Hartmann wavefront sensor (SHWFS) has found widespread applications in various fields, including atmosphere turbulence measurement,^{1,2} adaptive optics,^{3,4} laser beam quality testing,^{5,6} human eye aberration measurements,^{7,8} and communications.^{9,10}

The SHWFS operates on two fundamental procedures: wavefront slope calculation and wavefront reconstruction algorithms. The previous research on reconstruction algorithms primarily focused on spatial domain analysis^{11–14} to evaluate the performance of the reconstructors. Departing from conventional metrics, the spatial frequency response provides comprehensive insights into assessing distinct wavefront reconstructors. Several studies have explored frequency response analyses.

Bahk^{15,16} scrutinized the frequency response of three classical reconstructors—Hudgin,¹⁷ Fried,¹⁸ and Southwell.¹⁹ However, these analyses did not consider the low-pass effect of the microlens, limiting the frequency response to the reconstruction process rather than the entire wavefront sensing system.

Li *et al.*²⁰ proposed two weighted Fried reconstructors and examined the frequency response of the entire system. Nevertheless, the model’s complexity and the need to calculate weight values pose challenges.

This study introduces an enhanced reconstruction model based on Fan’s model. The frequency response of the wavefront sensing system for this proposed model is derived. In addition, an analysis is conducted to compare the reconstruction effectiveness of the proposed model to the Southwell model, one of the most successful classical models, across various frequency points.

The remainder of this paper is structured as follows: Sec. II presents the new geometry for the proposed reconstructor and outlines the derivation of the system frequency response formula. Section III encompasses the simulations conducted in this study. Section IV delves into the analysis of real experiments and reconstruction results. Subsequently, Sec. V offers discussions pertaining to the findings. Finally, Sec. VI presents the conclusions drawn from this study.

II. PRINCIPLE

A. Principle of the Fan's model reconstructor

The Shack–Hartmann wavefront sensor (SHWFS) consists of a camera and a microlens array, as shown in Fig. 1. The camera is located at the back focal plane of the microlens. In Fig. 1, the white peak points symbolize the subspots within the SHWFS. The centroid spot images, captured by the camera, depict the gradient of the incident wavefront phase through the observed centroid drifts.

Initially, the centroid coordinate can be expressed as

$$\begin{cases} x_c = \frac{\iint x I_i(x, y) dx dy}{\iint I_i(x, y) dx dy} \propto \iint \varphi_x dx dy, \\ y_c = \frac{\iint y I_i(x, y) dx dy}{\iint I_i(x, y) dx dy} \propto \iint \varphi_y dx dy, \end{cases} \quad (1)$$

where (x_c, y_c) signify the centroid coordinate, I_i represents the intensity at the pixel i , φ_x and φ_y are the wavefront phase gradients in the orthogonal directions, f denotes the focal length of the microlens array, and the integration region pertains to the corresponding sub-aperture.

Subsequently, the incident wavefront can be reconstructed by utilizing the measured wavefront slopes. The geometry of the proposed reconstructor is shown in Fig. 2, where the black dots denote locations where phase slopes are measured and the gray dots represent intermediate points. These gray dots are positioned at half steps in the “ x ” or “ y ” direction from the measured points. In contrast to the traditional Hudgin, Fried, and Southwell models, which underutilize slope information, the proposed reconstructor leverages five sample points of the slope for reconstruction measurement instead of the typical two. Hence, this reconstructor can be denoted as “Fan’s” model, similar to prior models.

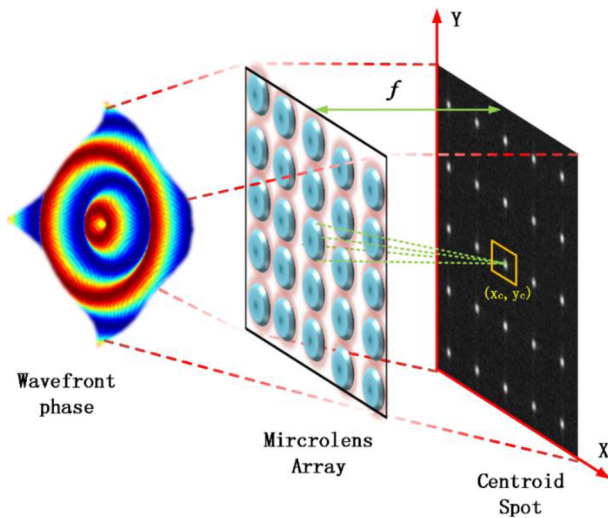


FIG. 1. SHWFS image model.

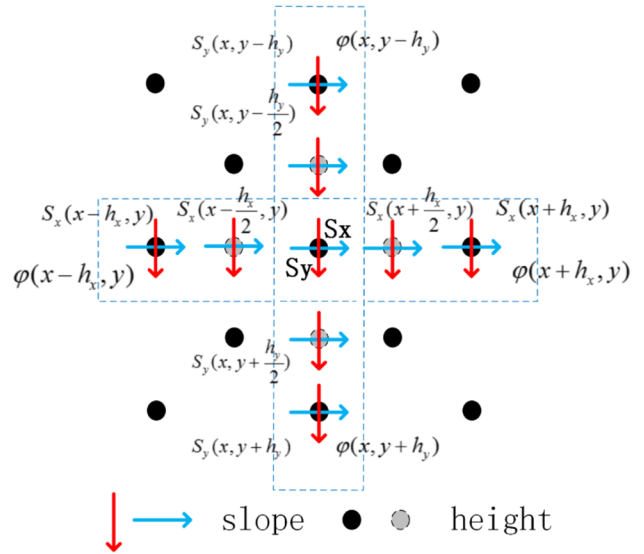


FIG. 2. Fan's reconstruction model geometry.

The finite difference methods can be used in spatial to describe the relationship between the measured slope S and the estimated phase $\hat{\varphi}$. The formula is

$$\sum_j a(j) S(i+j) = \sum_k b(k) \hat{\varphi}(i+k). \quad (2)$$

The suggested reconstruction model, the utilization of the five-order Newton–Cotes rule²¹ serves to augment the frequency response. The derived formula is represented as the following equation:

$$\begin{aligned} & \left[\frac{1}{90} \left[7S_x(x-h_x, y) + 32S_x\left(x - \frac{h_x}{2}, y\right) + 12S_x(x, y) \right. \right. \\ & \quad \left. \left. + 32S_x\left(x + \frac{h_x}{2}, y\right) + 7S_x(x+h_x, y) \right] \right. \\ & = \frac{1}{2h_x} [\varphi(x+h_x, y) - \varphi(x-h_x, y)] \\ & \left. \left[\frac{1}{90} \left[7S_y(x, y-h_y) + 32S_y\left(x, y - \frac{h_y}{2}\right) + 12S_y(x, y) \right. \right. \right. \\ & \quad \left. \left. + 32S_y\left(x, y + \frac{h_y}{2}\right) + 7S_y(x, y+h_y) \right] \right. \\ & = \frac{1}{2h_y} [\varphi(x, y+h_y) - \varphi(x, y-h_y)], \end{aligned} \quad (3)$$

where $S_x(x, y)$ and $S_y(x, y)$ are the x and y slopes, as shown in Fig. 2, and h_x and h_y are the x - and y -step sizes,

$$AS(x, y) = D\varphi(x, y) \quad (4)$$

$$A = \frac{1}{90} \begin{bmatrix} 7\delta(x - h_x, y) + 32\delta\left(x - \frac{h_x}{2}, y\right) + 12\delta(x, y) + 32\delta\left(x + \frac{h_x}{2}, y\right) + 7\delta(x + h_x, y) \\ 7\delta(x, y - h_y) + 32\delta\left(x, y - \frac{h_y}{2}\right) + 12\delta(x, y) + 32\delta\left(x, y + \frac{h_y}{2}\right) + 7\delta(x, y + h_y) \end{bmatrix}, \quad (5)$$

$$D = \frac{1}{2h_x h_y} \begin{bmatrix} h_y \delta(x + h_x, y) - h_y \delta(x - h_x, y) \\ h_x \delta(x, y + h_y) - h_x \delta(x, y - h_y) \end{bmatrix}. \quad (6)$$

It can be simplified as Eq. (4), where A is the space domain average operator, D is the space domain differential operator, and δ is the Dirac delta function.

B. Frequency domain S-H model

Considering the influence of the microlens array, the relationship between the actual slope and the measured slope can be expressed as

$$\begin{cases} S_x = \iint \hat{\phi}_x(x, y) \text{rect}\left(\frac{x - x_o}{d}\right) \text{rect}\left(\frac{y - y_o}{d}\right), \\ S_y = \iint \hat{\phi}_y(x, y) \text{rect}\left(\frac{x - x_o}{d}\right) \text{rect}\left(\frac{y - y_o}{d}\right), \end{cases} \quad (7)$$

where $\text{rect}(\cdot)$ is the rectangular window function, d is the width of the window, and (x_o, y_o) is the center coordinate of the selected sub-aperture.

Applying discrete Fourier transform (DFT) to Eq. (7), the frequency-domain formula is

$$\begin{cases} \tilde{S}_x = \tilde{\phi} \times \left[ik_x \text{sinc}\left(\frac{\omega_x}{2\pi}\right) \text{sinc}\left(\frac{\omega_y}{2\pi}\right) \right], \\ \tilde{S}_y = \tilde{\phi} \times \left[ik_y \text{sinc}\left(\frac{\omega_x}{2\pi}\right) \text{sinc}\left(\frac{\omega_y}{2\pi}\right) \right], \end{cases} \quad (8)$$

where $\tilde{S}_x$ and $\tilde{S}_y$ are the frequency-domain wavefront slopes in the orthogonal directions, $\tilde{\phi}$ is the DFT of the wavefront phase, k_x and k_y are the discrete spatial frequency in the orthogonal directions, and $\omega_x = k_x d$ and $\omega_y = k_y d$ are the frequency-domain parameters.

C. System frequency response function

The frequency response of Fan's reconstruction model is determined through the employment of the frequency domain least-squares method. According to Parseval's Theorem,²²

$$\varepsilon = \sum \{ |\tilde{D}_x \tilde{\phi} - A_x \tilde{S}_x|^2 + |\tilde{D}_y \tilde{\phi} - A_y \tilde{S}_y|^2 \}. \quad (9)$$

In Eq. (7), ε is the sum of the reconstruction calculation error, and $\tilde{D}_x$ and $\tilde{D}_y$ are the frequency domain differential operators, which can be described as

$$\tilde{D}_x = 1/2h_x [\exp(ik_x h_x) - \exp(-ik_x h_x)], \quad (10)$$

$$\tilde{D}_y = 1/2h_y [\exp(ik_y h_y) - \exp(-ik_y h_y)], \quad (11)$$

where h_x and h_y are the x - and y -step sizes, k_x and k_y are the spatial frequencies that range from $-\pi/h_x$ to $\pi/h_x - 2\pi/(Nh_x)$ and from $-\pi/h_y$ to $\pi/h_y - 2\pi/(Nh_y)$, N is the symbol of the sampling points, and the sampling interval is $2\pi/Nh_x$ and $2\pi/Nh_y$, respectively.

Here, $\tilde{A}_x$ and $\tilde{A}_y$ are the frequency-domain average operators, which can be described as

$$\tilde{A}_x = 1/90 [7 \exp(ik_x h_x) + 32 \exp(ik_x h_x/2) + 12 + 32 \exp(-ik_x h_x/2) + 7 \exp(-ik_x h_x)], \quad (12)$$

$$\tilde{A}_y = 1/90 [7 \exp(ik_y h_y) + 32 \exp(ik_y h_y/2) + 12 + 32 \exp(-ik_y h_y/2) + 7 \exp(-ik_y h_y)], \quad (13)$$

and $\tilde{\phi}$ is the estimated value of the frequency domain wavefront phase. So,

$$\tilde{\phi} = \frac{\tilde{D}_x^* \tilde{A}_x \tilde{S}_x + \tilde{D}_y^* \tilde{A}_y \tilde{S}_y}{|\tilde{D}_x|^2 + |\tilde{D}_y|^2}. \quad (14)$$

The whole system frequency response is

$$H_{\text{proposed}} = \frac{\tilde{\phi}}{\tilde{\phi}} = 4/45 \frac{\left\{ \omega_x \sin \omega_x (7 \cos \omega_x + 32 \cos(\omega_x/2) + 6) \right.}{\omega_x \omega_y (\sin^2 \omega_x + \sin^2 \omega_y)} \left. + \omega_y \sin \omega_y (7 \cos \omega_y + 32 \cos(\omega_y/2) + 6) \right\} * \sin(\omega_x/2) \sin(\omega_y/2), \quad (15)$$

where $\omega_x, \omega_y \in [-\pi, \pi]$, as shown in Fig. 3.

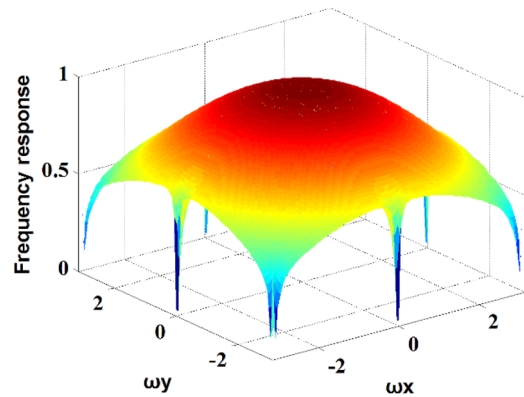


FIG. 3. Fan's reconstruction model for the 2D system frequency response.

It is important to note that the system response encounters a low-pass filter effect due to the microlens array. This system response function is both real and non-negative, encompassing frequency parameters spanning from $-\pi$ to π .

D. Noise error propagation

The noise error propagation curve illustrates the sensitivity of the reconstructed phase's noise to its input noise. The noise propagation coefficient for the proposed reconstructor is computed using frequency domain filter functions, as expressed in the following equation:

$$\sigma_{\varphi}^2 = \frac{1}{N_t} \sum \langle |\Delta\tilde{\varphi}|^2 \rangle, \quad (16)$$

where σ_{φ} is the root-mean-square of the reconstructed wavefront phase φ , N_t is number of sample points, and $\langle \cdot \rangle$ denotes the ensemble average of the quantity inside.

The power spectrum of the proposed reconstructor correlates with the absolute square of the linear system function, as delineated by linear stochastic system theory. So,

$$\langle |\Delta\tilde{\varphi}|^2 \rangle = \frac{(|D_x A_x \Delta\tilde{S}_x|^2 + |D_y A_y \Delta\tilde{S}_y|^2)}{(|D_x|^2 + |D_y|^2)^2}. \quad (17)$$

Assuming that $\Delta\tilde{S}_x$ and $\Delta\tilde{S}_y$ are uncorrelated white noise with the variance of σ_s^2 for each, the noise error propagation coefficient η is

$$\eta = \frac{\sigma_{\varphi}^2}{\sigma_s^2} = \frac{1}{M^2} \sum_{k_x, k_y} \frac{(|D_x A_x|^2 + |D_y A_y|^2)}{(|D_x|^2 + |D_y|^2)^2}, \quad (18)$$

where M is the root-mean-square of reconstructed wavefront phase.

III. SIMULATION

To assess the frequency domain performance of the proposed reconstructor, a series of wavefront phase images at varying frequencies are generated for reconstruction in simulations, as shown in Fig. 4. The formula is

$$z = 0.16\lambda \sin\left(2\pi \frac{L}{N} x \cdot T\right), \quad (19)$$

where $L = 30$ mm represents the spatial length of the wavefront phase image, $N = 4096$ denotes the spatial sampling numbers, x signifies the horizontal coordinate, $\lambda = 1053$ nm stands for the spatial wavelength, and $T = 2, 4, 8, 16$, and 32 indicates the number of periods. Corresponding to T , the normalized frequency is denoted as $f = 0.015625, 0.03125, 0.0625, 0.125$, and 0.25 , respectively.

These parameters characterize the original wavefronts at various frequencies. The parameter's value T ranges from low to high frequencies, which is a reasonable simulation value. The resulting reconstruction outcomes of the proposed reconstructor are shown in Fig. 5. The peak-valley (PV) and root mean square (rms) values for both the proposed reconstructor and the Southwell reconstructor are presented in Table I.

Both the proposed reconstructor and the Southwell reconstructor can reconstruct the original incident wavefront shape in simulations, but they exhibit larger errors in higher frequency regions. Simultaneously, the errors of the proposed reconstructor

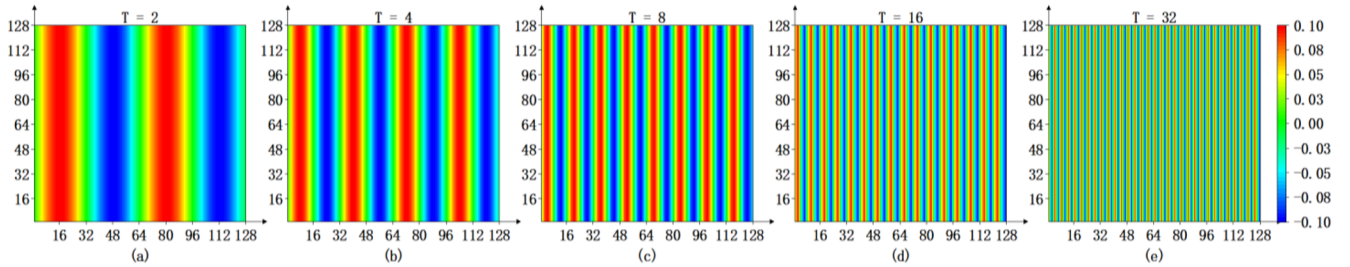


FIG. 4. Different periods' sinusoidal function wavefront phase original images simulation (a) $T = 2$, (b) $T = 4$, (c) $T = 8$, (d) $T = 16$, and (e) $T = 32$.

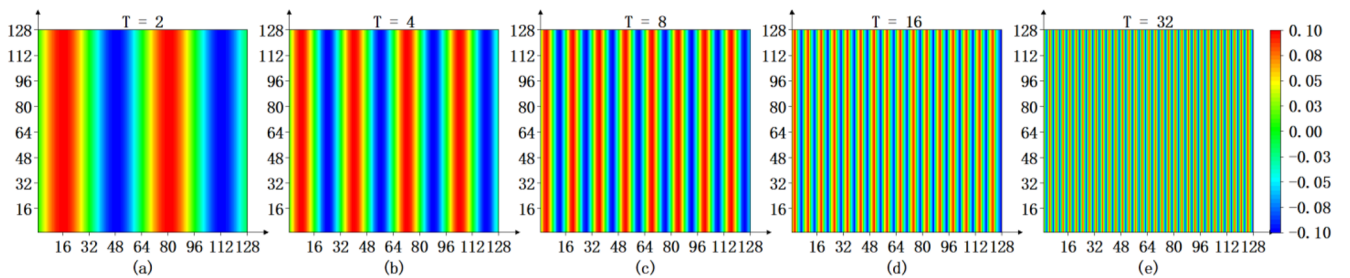


FIG. 5. Proposed model reconstruction wavefront effect (a) $T = 2$, (b) $T = 4$, (c) $T = 8$, (d) $T = 16$, and (e) $T = 32$.

TABLE I. Reconstruction computational time.

	Southwell	Proposed
Computational time	26.264 ms	29.555 ms

TABLE II. Reconstruction PV and rms. Boldface denotes that the proposed method has a smaller value than Southwell's model

T	PV (λ)		rms (λ)	
	Proposed	Southwell	Proposed	Southwell
2	0.0134	0.0136	0.0047	0.0048
4	0.0136	0.0146	0.0048	0.0052
8	0.0147	0.0185	0.0053	0.0067
16	0.0176	0.0320	0.0067	0.0122
32	0.0160	0.0591	0.0078	0.0288

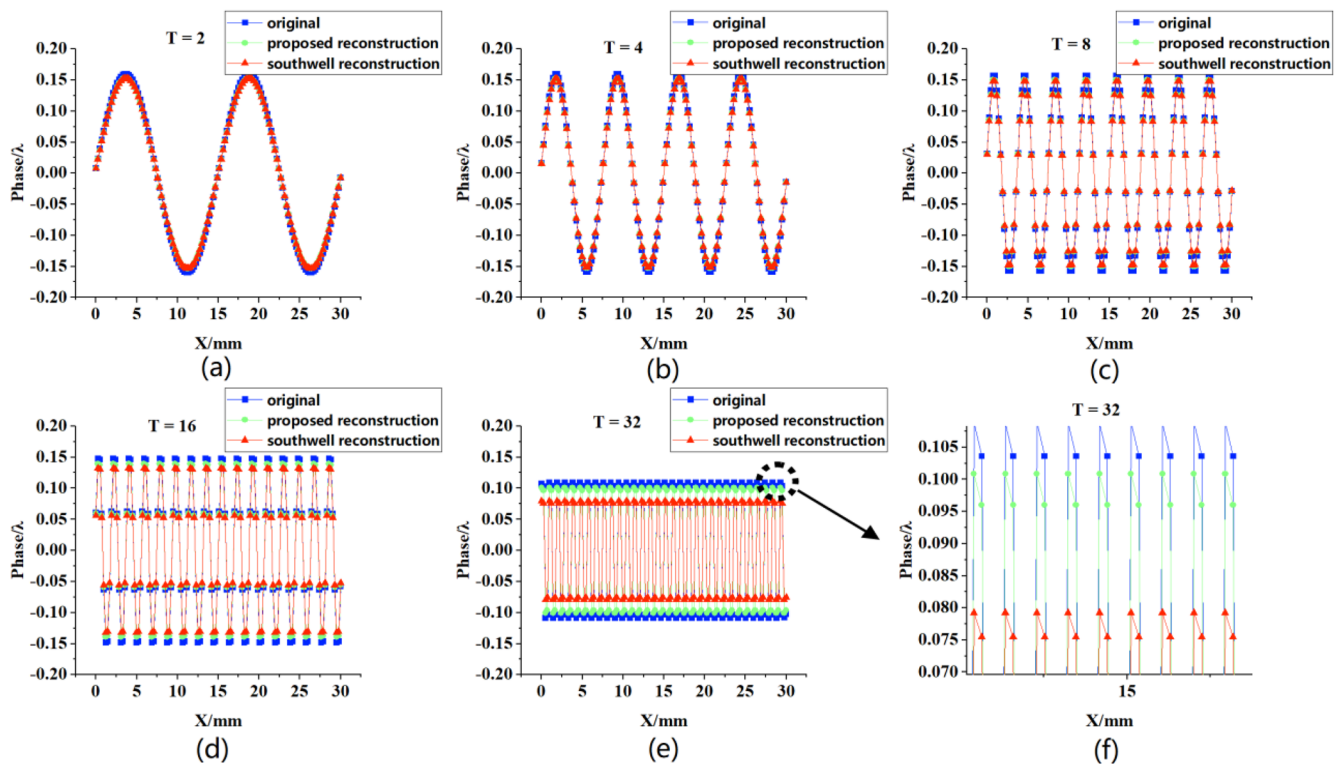
are consistently smaller than those of the Southwell reconstructor at each frequency point (Table II).

The simulated incident wavefront reconstruction results for the x axis are shown in Fig. 6. It is evident that both the reconstructors exhibit good performances in low-frequency regions, and the reconstruction results closely coincide with the original

wavefront. However, as the frequency increases, the reconstruction effect deteriorates.

When T is 32, the reconstructed phase peak value of the proposed method is 0.1008λ and the reconstructed phase peak value of Southwell method is 0.0796λ . [Fig. 6(e)] It indicates that the proposed Fan's model reconstructor performs better compared to the Southwell reconstructor in high-frequency regions.

Figure 7 shows the normalized frequency domain response curves of the proposed reconstructor and the Southwell reconstructor, represented by distinct colors. The green line represents the upper limit of the frequency response. The black line and magenta line represent the theoretical normalized frequency response characteristics of the Southwell model and the model proposed in this paper, respectively. The normalized frequency points selected for simulation are as follows: 0.015 625, 0.031 25, 0.0625, 0.125, 0.25, and 0.475. The blue circles represent the frequency response of the model proposed in this paper at the corresponding frequency points, while the red dots represent the frequency response of the Southwell model at the corresponding frequency points. The simulation experiment data closely align with the theoretical values. Notably, the proposed reconstructor exhibits a 27% enhancement in response bandwidth compared to the Southwell model. While minimal disparity is observed between the two reconstructors in low-frequency regions, the proposed reconstructor demonstrates notably higher response levels in mid- and high-frequency (Fig. 8).

**FIG. 6.** X-axis reconstruction results of the proposed model and the Southwell model (a) $T = 2$, (b) $T = 4$, (c) $T = 8$, (d) $T = 16$, (e) $T = 32$, and (f) subgraph of $T = 32$.

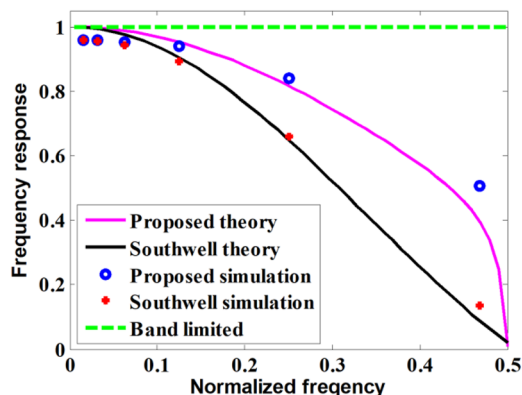


FIG. 7. 1D frequency response of the proposed model and the Southwell model.

IV. EXPERIMENT

To demonstrate the feasibility of the proposed reconstructor in actual measurement, the optical experiment is designed and shown in Fig. 9. The expanded beam from a 635 nm wavelength laser is used as the incident light. The collimating lens functions is used to generate collimated light. The aperture diaphragm controls the diameter of the circle spot. The polarizer is used to produce the polarized light, to which SLM has a good response effect. A series of sine masks are loaded on the SLM (UPOLabs HD5LM80R; pixel pitch is $8\ \mu\text{m}$, and resolution is 1920×1200) and placed on the focal plane of a 4f system. A self-developed Shack–Hartmann wavefront sensor (lenslet pitch = $236.8\ \mu\text{m}$, effective focal length = $6.775\ \text{mm}$, pixel size = $7.4\ \mu\text{m}$) is set in another focal plane of the 4f system. The camera is Vieworks VH-4MG2-M20. The list of camera parameters used is presented in Table III. The beam diameter is reduced to $7.5\ \text{mm}$ on the HS wavefront sensor and, equivalently, the wavefront sampling

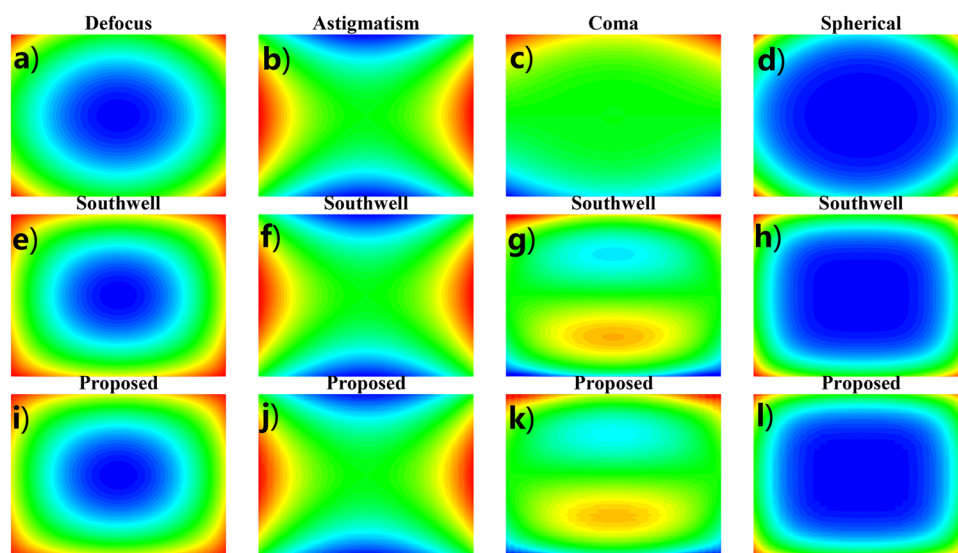


FIG. 8. 2D aberration reconstruction result of the proposed model and the Southwell model (simulation).

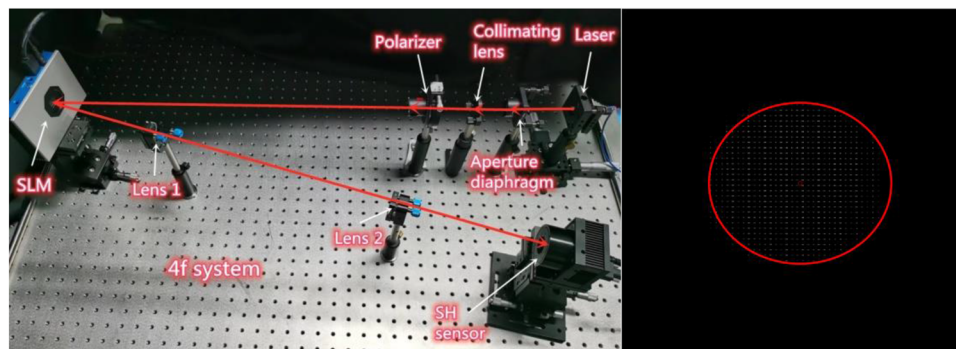


FIG. 9. Experimental optical path and effective reconstruction area.

TABLE III. Camera parameters.

Resolution	2048 × 2048	Frame rate	20 fps
Pixel size	7.4 μm	Shutter type	Global shutter
Pixel depth	12 bit	Mono/color	Mono

is done using 64×64 microlenses. Both the SLM and HS wavefront sensors are calibrated before the experiment to avoid the inherent phase variations.

Despite the somewhat suboptimal response linearity and the non-rectangular aperture shape, the proposed method successfully reconstructs the wavefront phase, as shown in Figs. 10(a)–10(d). Simultaneously, Figs. 10(e)–10(h) show the reconstruction results obtained using Southwell's method. The interpolation method used in false-color illustrations is bicubic interpolation. The 1D differences along the x axis between the two reconstructions are shown in Figs. 10(i)–10(l).

As shown in Fig. 10, both reconstructors effectively restore the original wavefront phase. Nonetheless, the proposed reconstructor exhibits greater response magnitude values compared to the Southwell reconstructor across various frequency points. Notably, the difference in response between the two reconstructors amplifies with increasing frequency. The heightened high-frequency response of the proposed reconstructor enables the retrieval of more detailed information concerning the incident wavefront.

V. DISCUSSION

The simulation experiment is performed using an Intel Xeon E5-2690 v4($\times 2$) CPU (2.6 GHz, 256 GB RAM) and MATLAB software (version R2014a, Math Works, USA). In the real experiment, although the resolution of the HS wavefront sensor is 64×64 , the actual effective area is 32×32 due to the pupil diameter, as shown in Fig. 9. However, this does not affect the experiment results. The reconstructed results are obtained after removing the effects of tilt and defocus.

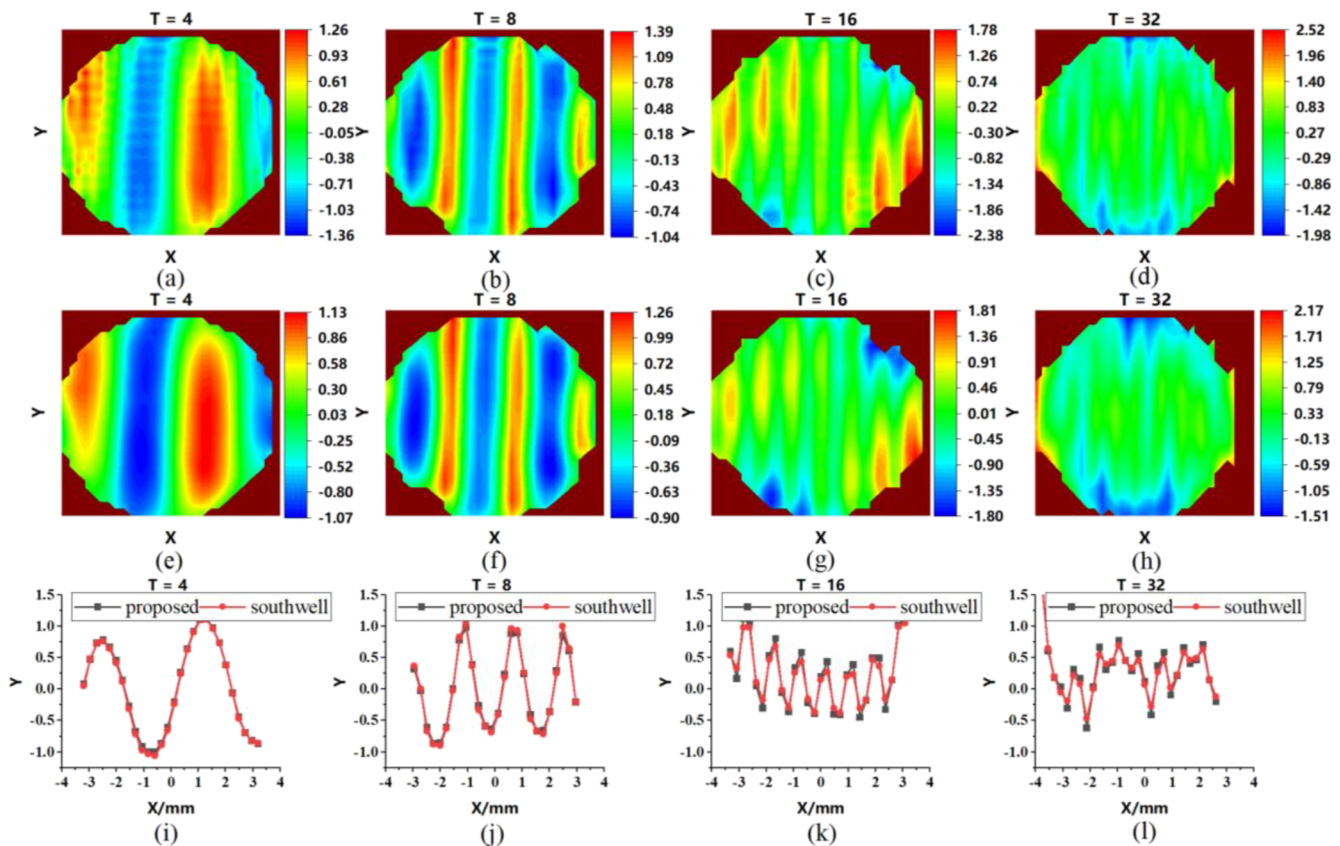


FIG. 10. Real experiment reconstruction results. (a) $T = 4$, proposed reconstruction; (b) $T = 8$, proposed reconstruction; (c) $T = 16$, proposed reconstruction; (d) $T = 32$, proposed reconstruction; (e) $T = 4$, Southwell reconstruction; (f) $T = 8$, Southwell reconstruction; (g) $T = 16$, Southwell reconstruction; (h) $T = 32$, Southwell reconstruction; (i) $T = 4$, x-axis reconstruction results of proposed and Southwell; (j) $T = 8$, x-axis reconstruction results of proposed and Southwell; (k) $T = 16$, x-axis reconstruction results of proposed and Southwell; and (l) $T = 32$, x-axis reconstruction results of proposed and Southwell.

The proposed algorithm also has limitations, including high computational complexity, sensitivity to noise, and boundary effects. The proposed algorithm needs to gradually approximate the true phase of the wavefront through an iterative process, which may lead to a large amount of computation, especially when dealing with large-scale datasets. If the measurement data contain noise, it may be amplified during the iteration, affecting the quality of the final reconstruction result. When dealing with wavefronts with complex boundaries, the proposed algorithm may be difficult to accurately process the boundary conditions, resulting in the deterioration of reconstruction accuracy near the boundary.

The proposed reconstructor efficiently reconstructs the original wavefront shape and details across different periods of the sinusoidal wavefront in the spatial domain. The new method, utilizing five points of slope data, enhances the bandwidth of the frequency response. Exploring the incorporation of more slope data could be a potential avenue for future work.

VI. CONCLUSION

In summary, this study presents a zonal reconstructor built upon Fan's model, aiming to improve the wavefront reconstruction performance in mid- and high-frequency ranges. The formulated equation for the proposed reconstructor closely aligns with theoretical frequency response curves, as confirmed through simulation experiments. Notably, the suggested reconstructor demonstrates a 27% expansion in the bandwidth area compared to the Southwell reconstructor. Real-world experiments further validate the superior performance of the proposed reconstructor, particularly in efficiently addressing high-frequency regions.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yao Fan: Conceptualization (lead); Data curation (lead); Methodology (lead); Software (lead); Writing – original draft (lead). **Yaxuan Duan:** Conceptualization (supporting); Funding acquisition (lead). **Zhengshang Da:** Project administration (equal); Resources (equal). **Yang Yue:** Supervision (supporting); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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